Sample output :

```
tharunika@localhost:~
tharunika@localhost:~                tharunika@localhost:~
[tharunika@localhost ~]$ nslookup google.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.122.139
Name:   google.com
Address: 142.250.122.113
```

```
tharunika@localhost:~
tharunika@localhost:~                tharunika@localhost:~
[tharunika@localhost ~]$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         _gateway        0.0.0.0         UG    100    0        0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U     100    0        0 enp0s3
[tharunika@localhost ~]$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost.localdo:38604 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:54522 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:38620 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:49992 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:54534 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:49904 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:54548 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:54512 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:38600 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:38630 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:49918 101.6.15.130:http       TIME_WAIT
udp        0      0 localhost.localdo:bootpc _gateway:bootps        ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ]         DGRAM      CONNECTED     22818    /run/chrony/chronyd.s
ock
unix  2      [ ]         DGRAM                    28072    /run/user/1000/system
d/notify
[tharunika@localhost ~]$ getmac
bash: getmac: command not found...
[tharunika@localhost ~]$ arp -a
? (gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[tharunika@localhost ~]$ ip neigh
10.0.2.2 dev enp0s3 lladdr 52:54:00:12:35:02 STALE
[tharunika@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[tharunika@localhost ~]$
```

```
tharunika@localhost:~
tharunika@localhost:~                tharunika@localhost:~
[tharunika@localhost ~]$ ip neigh
10.0.2.2 dev enp0s3 lladdr 52:54:00:12:35:02 STALE
[tharunika@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[tharunika@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.1.1
[tharunika@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[tharunika@localhost ~]$ ss -tuln
Netid  State   Recv-Q  Send-Q    Local Address:Port     Peer Address:Port
udp    ESTAB   0       0       10.0.2.15%enp0s3:68          10.0.2.2:67
[tharunika@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
```

Exp NO:1

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEM

AIM:

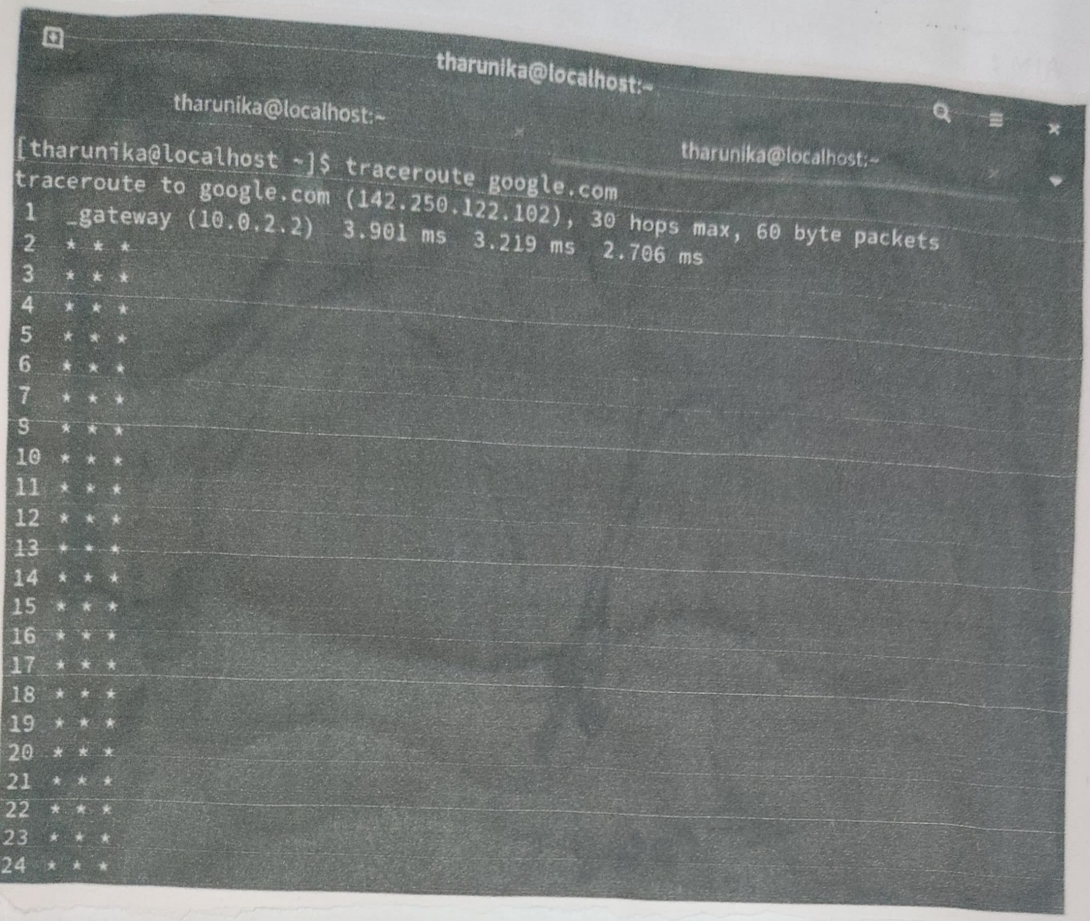
To study & execute basic networking commands in window & Linux operating system for viewing network configuration, testing connectivity, checking routine information, monitoring network status, troubleshooting

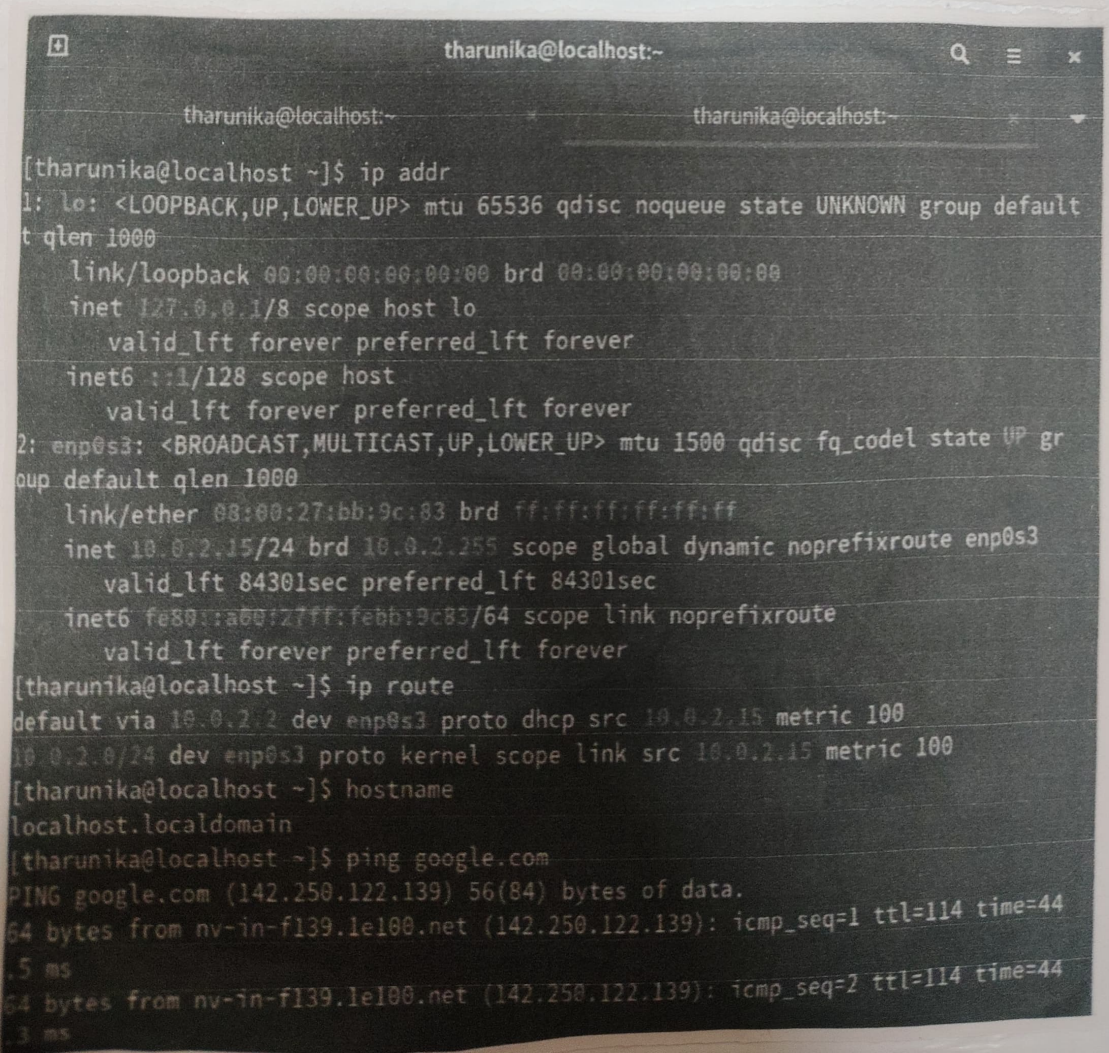
CODE:

Open commands prompts

commands

```
ipconfig
ipconfig /all
ping google.com
tracert google.com
nslookup google.com
route print
netstat -a
netstat
get-mac
hostname
arp -a
```


tharunika@localhost:~
tharunika@localhost:~
tharunika@localhost:~
[tharunika@localhost ~]$ traceroute google.com
traceroute to google.com (142.250.122.102), 30 hops max, 60 byte packets
1 _gateway (10.0.2.2) 3.901 ms 3.219 ms 2.706 ms
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
7 * * *
8 * * *
9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *


tharunika@localhost:~
tharunika@localhost:~
tharunika@localhost:~
[tharunika@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 84301sec preferred_lft 84301sec
inet6 fe80::a00:27ff:febb:9c83/64 scope link noprefixroute
valid_lft forever preferred_lft forever
[tharunika@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[tharunika@localhost ~]$ hostname
localhost.localdomain
[tharunika@localhost ~]$ ping google.com
PING google.com (142.250.122.139) 56(84) bytes of data.
64 bytes from nv-in-f139.1e100.net (142.250.122.139): icmp_seq=1 ttl=114 time=44
.5 ms
64 bytes from nv-in-f139.1e100.net (142.250.122.139): icmp_seq=2 ttl=114 time=44
.3 ms

Open Linux Networking

Commands

ip addr

ip config

ip route

hostname

ping google.com

traceroute google.com

sudo apt install traceroute

nslookup google.com

sudo apt install dnsutils

net stat - a

arp -a

ip link

cat /etc/resolv.conf

ip link show

ss - tuln

curl google.com

RESULT :

Thus the window commands are Runed in command prompt & Expected output are observed, & the Linux commands are Runed through ubuntu & Expected output is Verified